Chapter 2.

Phonology and phonetic transcription

Careful speech (citational) vs. connected (conversational or colloquial) speech

Say the following expressions in both ways and observe differences in your speech:

Did you eat yet?

I got you.

=> *To describe what is happening here, we need phonetic transcription.*

Phonetics vs. Phonology? (p.35)

- Phonology is **the description of the systems and patterns of sounds** that occur in a language.

- It involves studying a language to determine **the distribution of sounds in words**, that is, we must discover **when a phonetic contrast is distinctive**, that is, when the sounds in contrast convey a difference in meaning and when a phonetic contrast does not convey a difference in meaning.

Phonetics vs. Phonology? (p.35)

- Children have to do this when they are learning to speak. They may not realize at first that, for example, there is a difference between the consonants at the beginning of words such as ‘white’ and ‘right.’

/waɪt/ /ɹaɪt/

Distinctiveness or contrast (p.36)

- When two sounds can be used to differentiate words, they are said to be distinct, or contrastive. The phonetic difference is phonemic - as opposed to a simple phonetic difference.

‘pill’ vs. ‘lip’ ‘right’ vs. ‘light’

- In phonetics and phonology, the following three terms are used synonymously - **distinctive, contrastive, phonemic**.

Phonetic details of sounds

Sound quality may change depending on phonetic environments: **a phoneme** may have **different allophones**

Q1. Words below all include /t/. (True or False?)

Q2. Words below all include [t]. (True or False?)

(a) time, retail, notorious

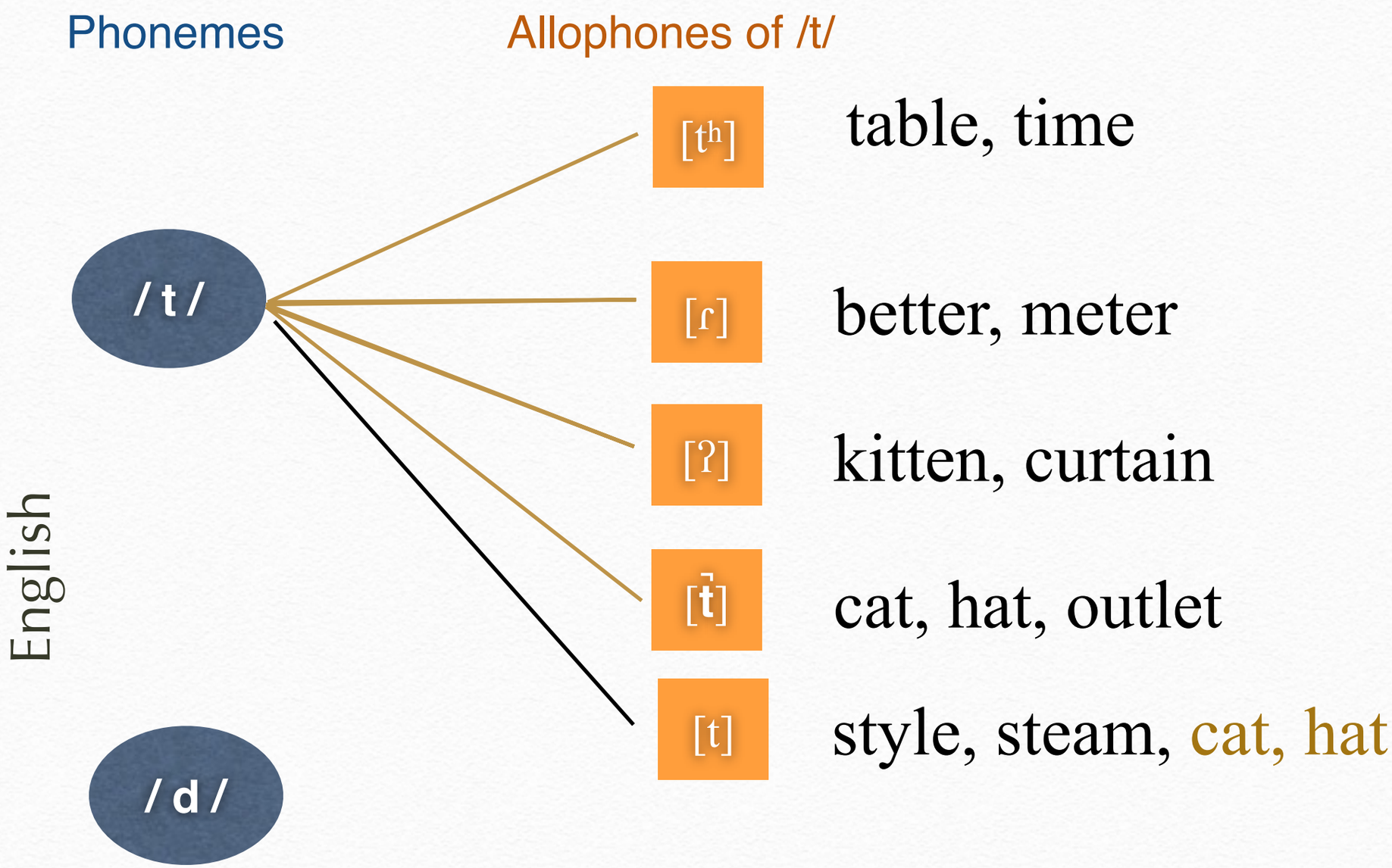
(b) star, stereo, steam

(c) butter, water, heater

(d) curtain, mountain, Manhattan

(e) set back, outlet, Batman

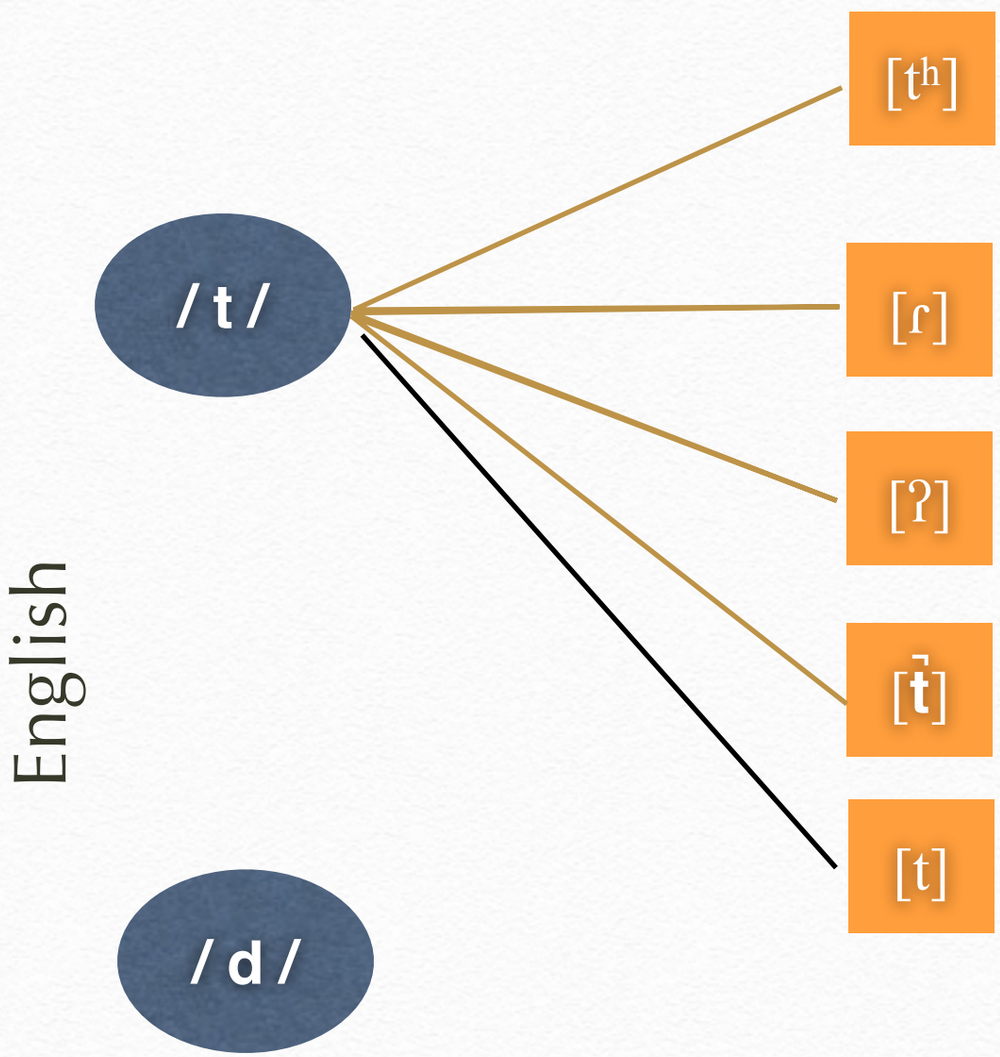
Phonetic details of sounds



Phonetic details of sounds

Phonemes

Allophones of /t/



English

a. /t/ becomes aspirated when it is in word-initial or in the onset of a stressed syllable except after /s/.

b. /t/ becomes a tap in the onset of an unstressed syllable intervocalically .

c. /t/ may become a glottal stop in the onset of an unstressed syllable when the unstressed vowel precedes a nasal.

d. /t/ is unreleased word finally or before another consonant

e. Elsewhere

Transcriptions of sounds

a. **Broad** transcription: *Transcribe mainly the significant articulations*

b. **Narrow** transcription: *Transcribe* ***more details*** *of a sound*

E.g.)	*'tie'*	*'Grand Canyon'*	
	/taɪ/	/kɛnjən/	
	[taɪ]	[kɛnjən]	Broad transcription
	[tʰaɪ]	[kʰɛ̃njən]	Narrow transcription
		[kʰɛ̃ɲən]	

Could be narrower

Transcriptions of sounds

a. **Phonemic** transcription: *Transcribe phonemic contrasts*

b. **Phonetic** transcription: *Transcribe phonetic details*

E.g.)	*‘team‘*	*‘steam’*	*‘heater’*	*‘written’*
/ /	/tim/	/stim/	/hitəɹ/	/rɪtən/
[ ]	[tʰim]	[stim]	[hirɚ]	[ɹɪʔn̩]

Example [1] Mary-merry-marry merger

"Mary, dear, make me merry; say you'll marry me."

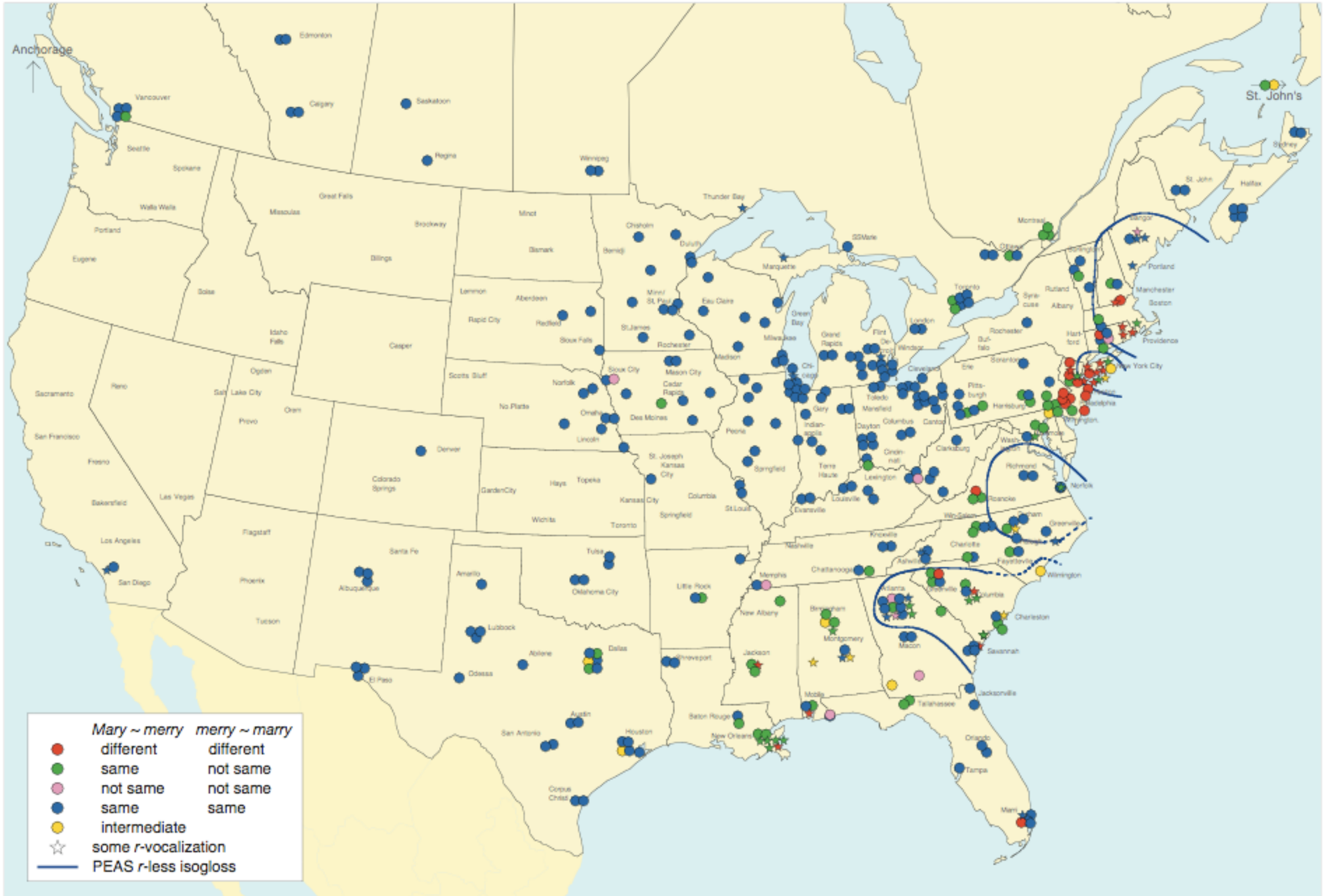
Mary [mɛəɹi]

merry [mɛɹi]

marry [mæɹi]

} [ɛ] Northern America (Southern including Baltimore and Delaware)

cf. Philadelphia: three way distinction but 'merry' sounds like 'murray'



Map 8.4. The merger of /ey, e, æ/ before intervocalic /r/

The great majority of North Americans pronounce *Mary*, *merry*, and *marry* as the same. A contrast of all three is maintained in the Mid-Atlantic states. *Merry* and *marry* are kept apart by a fair number of speakers in New England and the South as well as in Montreal, Quebec.

The Atlas of North American English (Labov et al, 2006)

Example [2] ‘writer’ vs. ‘rider’

In many dialects of American English

/ɹaɪtəɹ/	/ɹaɪdəɹ/
[ɹaɪɾɚ]	[ɹaɪɾɚ]

same pronunciation
(neutralized)

Canadian vowel raising: [aɪ] ~ [ʌɪ]

e.g., pipe, spice, tight, etc. [ʌɪ]

/aɪ/ e.g., vibe, spider, tide, etc. [aɪ]

writer [ʌɪ] vs. rider [aɪ]

Generalisation?

PHONOLOGY (Read together p.47-48)

sit vs. shit pie vs. spy easier, leisure, age

Phoneme

Allophone

Sound A	[s]	[t]
Sound B	[ʃ]	[tʰ]

Contrastive

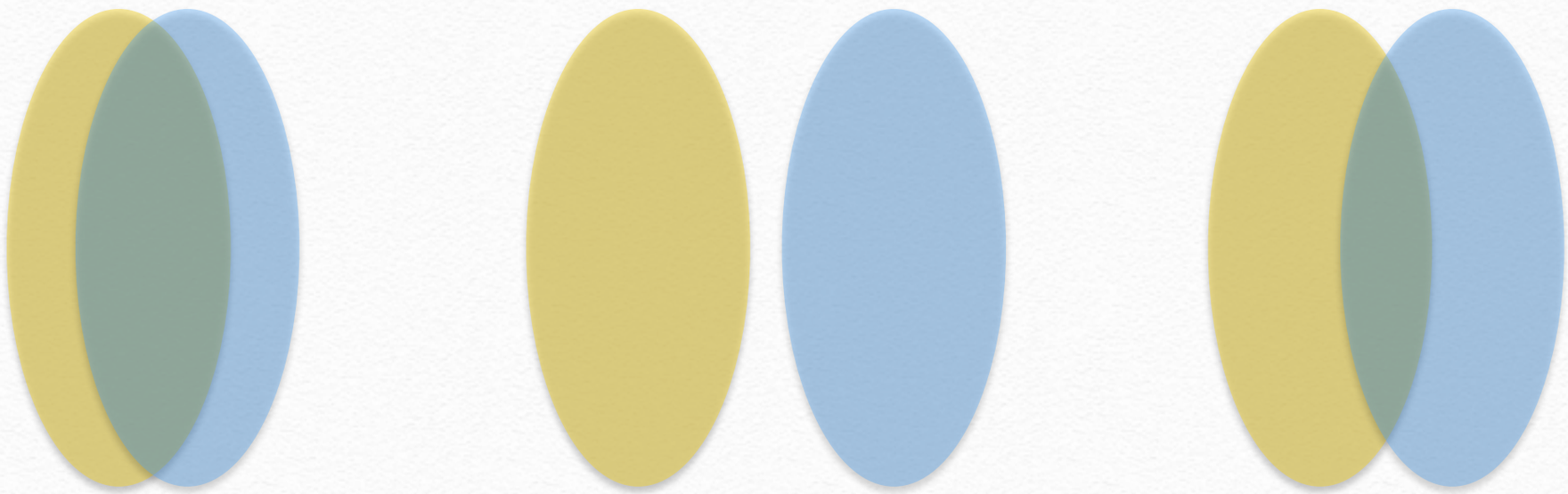
Complementary

Sound distribution

Free variation

Sounds and their distribution (p.49)

sit vs. shit peak vs. speak easier, leisure, age



contrastive
distinctive
phonemic

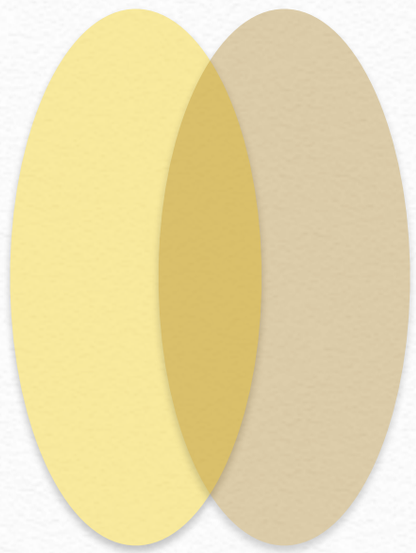
Complementary

contrastive
distinctive
phonemic

Free variation

‘a (book)’ ‘neither’ ‘economic’

‘either’



Distributions can overlap
but there is no meaning difference.

Diacritics

aspirated (consonant)	Cʰ	e.g., captain [kʰæp̄tən],
syllabic consonant	ç̩	e.g., little [lɪr̩]
vowel length	vː	e.g., cap vs. cab
Unreleased	c̄	e.g., captain [kʰæp̄ʔn̩]
Unvoiced	ç̥, v̥	e.g., pr̲ay [p̥ɹeɪ], pl̲ay [p̥leɪ]
nasalized	ṽ	e.g., ca̲nyon [æ̃]
velarized, dentalized	c̪	e.g., hea̲lth [ɫ̪]

More diacritics

ɚ vowel + r ('r-coloured vowel' when unstressed)

e.g., pray<u>er</u>

ɝ vowel + r ('r-coloured vowel'when stressed)

e.g., h<u>er</u>b

Practice #1

Reading transcription

1. [pʰəɹeɪd]
2. [bʌɾə˞]
3. [ʃɪɾi]
4. [ʤɛmd]
5. [beɪʒ]
6. [pʰeɪdʒ]
7. [dɪkʰoʊd]

정답: 1. parade, 2. butter, 3. shitty, 4. jammed, 5. beige, 6. page, 7. decode

Practice #2

Transcription of words

A

B

1. peanut

2. boy

3. tree

4. moment

5. caught

6. rain

7. total

Phonemic transcription on dictionaries: www.dictionary.com [show IPA]